

K. J. Somaiya School of Engineering

Somaiya Vidyavihar University

TY BTech AI&DS A3
OOSE Lab Mini Project

on

KJS-CES-01:
AI-Based Climate Intelligence
System for Heatwave Monitoring,
Prediction, and Early Warning.

Course : Object Oriented Software Engineering (OOSE)

Prepared By : Vedang Kulkarni & Rajat Kumar

Roll Numbers : 16014224053 & 16014224054

Under Guidance of : Prof. Deepti Patole

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Chapter 1

Project Planning

1.1 Problem Definition

Traditional heatwave monitoring systems rely mainly on meteorological parameters such as temperature, humidity and heat index to detect heatwaves. While these systems accurately monitor climatic conditions, they fail to capture the real-time impact on people. Citizens often post about extreme heat, dehydration, power cuts, health issues and outdoor difficulties on social media before official reports become available. Since existing systems ignore this valuable real-time information, authorities may respond late to emerging situations.

Therefore, an intelligent software system is required that integrates meteorological datasets with live social media sentiment analysis using Artificial Intelligence to generate faster heatwave alerts, monitor public distress and support timely decision-making for disaster management agencies.

1.2 Project Scope

1.2.1 Modules Included

- User Registration & Login
- Weather Data Collection Module
- Social Media Data Collection Module
- AI-based Sentiment Analysis
- Heatwave Severity Prediction
- Alert Generation
- Dashboard Visualization

- Historical Reports
- Admin Panel

1.2.2 Modules Excluded

- Satellite Image Processing
- Hardware Sensor Deployment
- Ambulance Dispatch
- Hospital Management
- Emergency Call Services
- Multi-language NLP beyond English

1.2.3 Assumptions

- Weather APIs provide accurate data.
- Social media APIs are accessible.
- Internet connectivity is available.
- AI model has sufficient training data.

1.2.4 Constraints

- API request limits
- Privacy regulations
- Noisy social media data
- Real-time processing requirements
- Limited computing resources

1.3 Choice of Process Model

The project follows the Agile Scrum software development model.

Agile is the most suitable process model because AI applications continuously evolve with new datasets, changing user requirements and model improvements. The project requires frequent testing of machine learning models, integration of new APIs and regular feedback from users. Scrum allows development in short iterations, making it easy to improve prediction accuracy, dashboard features and alert mechanisms while maintaining software quality.

1.4 Roles and Responsibilities

Role	Responsibilities
Project Manager	Planning, scheduling, resource allocation and project monitoring
Business Analyst	Requirement gathering and stakeholder communication
UI/UX Designer	Designing wireframes, dashboard and user interface
Backend Developer	API integration, database management and business logic
AI/ML Engineer	Sentiment analysis model and heatwave prediction model
Frontend Developer	Dashboard implementation and user interaction
Database Administrator	Database design, optimization and security
Tester (QA Engineer)	Functional testing, bug reporting and performance testing

Table 1.1: Project Roles and Responsibilities

1.5 GUI-Based Implementation

The Dashboard module provides real-time visualization of weather conditions, AI-based heatwave prediction, public sentiment trends, alerts, and historical reports. It enables administrators and disaster management authorities to monitor heatwave severity and take timely preventive actions.

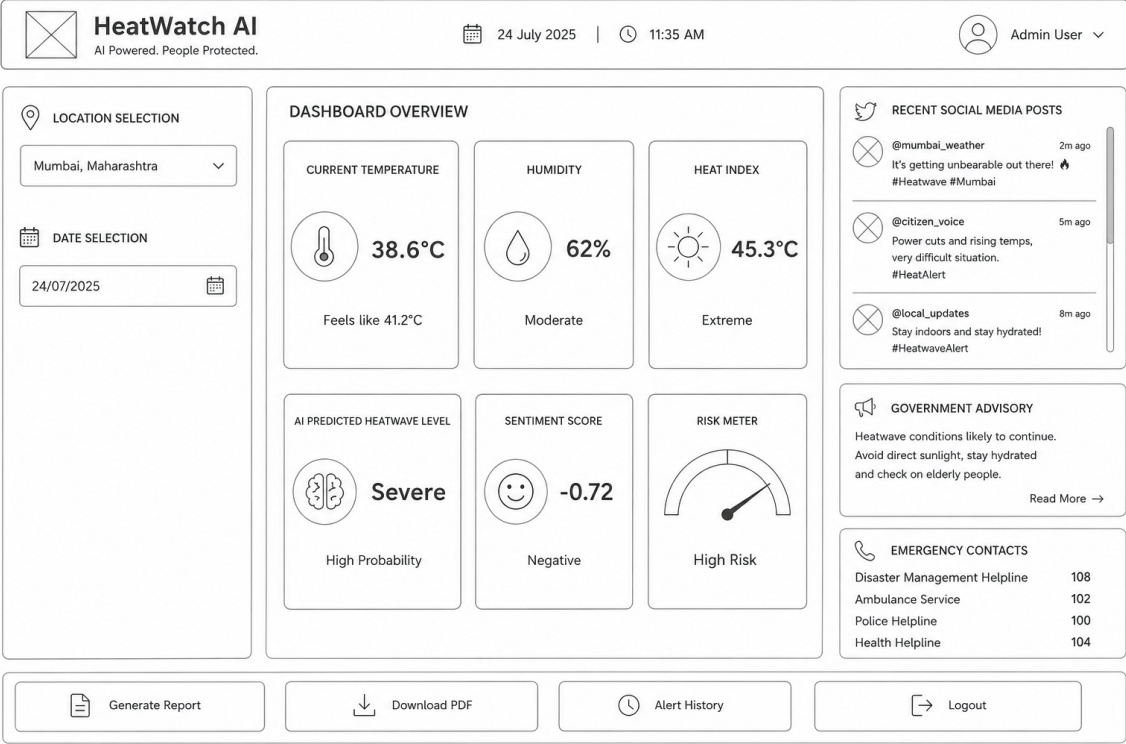


Figure 1.1: HeatWatch Dashboard GUI (Wireframe Model)

Chapter 2

Software Requirement Specification

2.1 Product Overview

The proposed software product, **HeatWatch AI**, is an AI-Based Climate Intelligence System for Heatwave Monitoring, Prediction, and Early Warning. The primary objective of the system is to provide timely and intelligent heatwave alerts by integrating meteorological data with Artificial Intelligence-based social media sentiment analysis.

The system continuously collects weather information from trusted weather APIs and combines it with publicly available social media posts to identify signs of heat-related public distress. Machine Learning models analyse both sources of information to predict heatwave severity, estimate potential risks, and generate early warnings for disaster management authorities.

The application will be deployed as a web-based platform that can be accessed using modern web browsers over an internet connection. The system is intended to support government agencies, disaster management authorities, environmental researchers, and system administrators in monitoring climate conditions and making informed decisions.

The intended users include:

- Disaster Management Officers responsible for monitoring and responding to heat-wave situations.
- Government Authorities responsible for planning emergency response strategies.
- Environmental Researchers interested in analysing climatic trends.
- System Administrators responsible for managing users, APIs, and system maintenance.

The expected users possess basic computer literacy and experience using web applications. No specialized programming knowledge is required for operating the system.

2.2 Specific Requirements

The Software Requirement Specification defines the functional and non-functional requirements of the proposed HeatWatch AI system. These requirements provide sufficient detail for software designers, developers, testers, and stakeholders to understand, implement, verify, and validate the system.

The requirements described in this chapter are externally observable and measurable, ensuring that every feature can be tested during system verification.

2.3 External Interface Requirements

The HeatWatch AI system interacts with users, external software services, communication networks, and hardware devices through well-defined interfaces.

2.3.1 User Interfaces

The HeatWatch AI system shall provide a responsive and user-friendly graphical interface that allows users to easily monitor climate conditions and heatwave predictions.

The user interface shall include the following components:

- Secure Login and Registration page.
- Dashboard displaying current weather information.
- AI-generated heatwave severity prediction.
- Social media sentiment analysis visualization.
- Interactive charts and graphs showing climatic trends.
- Alert notification panel for high-risk situations.
- Historical report generation and download facility.
- Administrative panel for user and system management.

The interface shall satisfy the following usability requirements:

- Simple and intuitive navigation.
- Responsive design supporting desktop and laptop devices.
- Clearly labelled menus and buttons.
- Consistent colour scheme representing risk levels.

- Minimal user input required for generating reports.
- Error messages should be meaningful and guide users toward corrective actions.

2.3.2 Hardware Interfaces

The proposed system does not require any dedicated hardware components for its normal operation.

The software shall execute on standard desktop or laptop computers with internet connectivity.

Minimum hardware requirements include:

- Dual-core processor or higher.
- Minimum 4 GB RAM.
- Minimum 10 GB available storage.
- Stable internet connection.
- Standard keyboard, mouse and display.

No specialized sensors or embedded hardware are required since weather information is obtained from online APIs.

2.3.3 Software Interfaces

The HeatWatch AI system communicates with multiple software services to perform its operations.

The major software interfaces include:

The communication between these software components shall use REST APIs and JSON formatted data wherever applicable.

2.3.4 Communication Protocols

The HeatWatch AI system shall support secure communication between clients, servers, and external APIs.

The communication requirements are listed below:

- HTTP/HTTPS protocol for web communication.
- RESTful API architecture for external services.
- JSON as the standard format for data exchange.

Software	Purpose
Operating System (Windows/Linux)	Executes the application and provides runtime environment.
Python 3.x	Backend programming language.
Flask / Django	Web application framework for backend services.
MySQL	Stores user information, weather data, alerts, and reports.
Weather API (OpenWeatherMap or similar)	Provides real-time meteorological data.
Social Media API	Retrieves public posts for sentiment analysis.
Machine Learning Libraries (Scikit-learn, TensorFlow)	Performs heatwave prediction and sentiment classification.
Matplotlib / Plotly	Generates graphs and dashboard visualizations.

Table 2.1: Software Interfaces

- Secure API authentication using API keys or OAuth where applicable.
- TCP/IP networking for internet communication.
- SSL/TLS encryption for secure transmission of sensitive information.

The system shall validate all incoming API responses before processing them to ensure data integrity and reliability.

2.4 Software Product Features

This section describes the major functional requirements of the proposed AI-Based Climate Intelligence System for Heatwave Monitoring, Prediction, and Early Warning. Each feature specifies the expected behaviour of the software in response to user inputs or system events.

2.4.1 Feature 1: User Registration and Authentication

Priority: Essential

Description

The system shall provide a secure registration and authentication mechanism for all authorized users.

Input

- Username
- Email ID
- Password

- User Role

Output

- Successful registration confirmation
- Secure user login
- Authentication error message for invalid credentials

System Behaviour

- The system shall validate all user inputs.
- The system shall encrypt user passwords before storing them.
- The system shall prevent duplicate user registrations.
- The system shall redirect authenticated users to the dashboard.

2.4.2 Feature 2: Weather Data Collection

Priority: Essential

Description

The system shall automatically retrieve real-time weather information using external weather APIs.

Input

- Selected location
- API request

Output

- Temperature
- Humidity
- Heat Index
- Wind Speed
- Weather Condition

System Behaviour

- The system shall connect to the configured Weather API.
- The system shall validate received data.
- The system shall update weather information periodically.
- The system shall notify the user if API communication fails.

2.4.3 Feature 3: Social Media Data Collection

Priority: Essential

Description

The system shall collect publicly available social media posts related to heatwave events.

Input

- Keywords
- Location
- Date Range

Output

- Retrieved social media posts
- Metadata associated with each post

System Behaviour

- The system shall connect to supported social media APIs.
- The system shall remove duplicate records.
- The system shall ignore incomplete or corrupted data.

2.4.4 Feature 4: AI-Based Sentiment Analysis

Priority: Essential

Description

The system shall analyse collected social media posts using Natural Language Processing techniques.

Input

- Social media posts

Output

- Positive sentiment
- Neutral sentiment
- Negative sentiment

- Sentiment score

System Behaviour

- The system shall preprocess textual data.
- The system shall classify user sentiments.
- The system shall store sentiment analysis results.

2.4.5 Feature 5: Heatwave Severity Prediction

Priority: Essential

Description

The system shall predict heatwave severity using machine learning models.

Input

- Weather parameters
- Sentiment scores
- Historical weather data

Output

- Low Risk
- Moderate Risk
- High Risk
- Predicted Severity Score

System Behaviour

- The system shall process input features using trained ML models.
- The system shall generate prediction confidence.
- The system shall update prediction results whenever new data becomes available.

2.4.6 Feature 6: Alert Generation

Priority: Essential

Description

The system shall generate early warnings whenever high heatwave risk is detected.

Input

- Heatwave prediction
- Severity threshold

Output

- Alert notification
- Warning message
- Risk Level

System Behaviour

- The system shall compare prediction values with predefined thresholds.
- The system shall generate alerts automatically.
- The system shall record every generated alert.

2.4.7 Feature 7: Dashboard Visualization

Priority: Important

Description

The system shall provide an interactive dashboard for monitoring climate information.

Input

- User selection
- Weather information
- Prediction results
- Sentiment statistics

Output

- Charts
- Heat maps

- Trend graphs
- Risk indicators

System Behaviour

- The system shall display real-time visualizations.
- The dashboard shall automatically refresh when new data becomes available.
- Users shall be able to filter information by location and date.

2.4.8 Feature 8: Historical Report Generation

Priority: Important

Description

The system shall generate downloadable historical reports.

Input

- Selected date range
- Selected location

Output

- PDF Report
- CSV Report
- Statistical Summary

System Behaviour

- The system shall retrieve historical records.
- The system shall generate formatted reports.
- Users shall be able to download reports.

2.4.9 Feature 9: Admin Panel

Priority: Important

Description

The system shall provide administrative functionalities.

Input

- Administrator credentials
- Management commands

Output

- Updated user information
- Updated system settings
- Activity logs

System Behaviour

- The administrator shall manage users.
- The administrator shall configure APIs.
- The administrator shall monitor system activities.
- The administrator shall maintain system logs.

2.4.10 Feature 10: Error Handling and Recovery

Priority: Important

Description

The system shall detect and recover from operational failures.

Input

- API failure
- Invalid user input
- Database failure
- Network interruption

Output

- Error notification

- Recovery status
- System log entry

System Behaviour

- The system shall validate all user inputs.
- The system shall retry failed API requests.
- The system shall log all critical errors.
- The system shall recover gracefully without data loss whenever possible.

2.4.11 Summary of Functional Requirements

ID	Requirement	Priority
FR-01	User Registration and Login	Essential
FR-02	Weather Data Collection	Essential
FR-03	Social Media Data Collection	Essential
FR-04	AI-based Sentiment Analysis	Essential
FR-05	Heatwave Severity Prediction	Essential
FR-06	Alert Generation	Essential
FR-07	Dashboard Visualization	Important
FR-08	Historical Report Generation	Important
FR-09	Administrative Panel	Important
FR-10	Error Handling and Recovery	Important

Table 2.2: Summary of Functional Requirements

2.5 Software System Attributes

The quality attributes of the HeatWatch AI system define the non-functional requirements that ensure the software performs efficiently, securely, and reliably under different operating conditions.

2.5.1 Reliability

The HeatWatch AI system shall provide reliable monitoring and prediction services by ensuring consistent operation during normal and peak usage conditions.

The reliability requirements are as follows:

- The system shall maintain an operational accuracy of at least 95% for weather data retrieval.

- The AI prediction module shall produce consistent prediction results using validated machine learning models.
- The system shall recover gracefully from temporary API failures without affecting stored data.
- The system shall automatically retry failed API requests up to three times before reporting an error.
- The system shall maintain detailed error logs for troubleshooting and system maintenance.

2.5.2 Availability

The HeatWatch AI system shall remain available to authorized users whenever internet connectivity and external APIs are operational.

Availability requirements include:

- The system shall maintain an availability of at least 99% during normal operation.
- The application shall automatically reconnect to external APIs after temporary communication failures.
- Previously stored historical data shall remain accessible even if external APIs become unavailable.
- User sessions shall remain active during temporary network interruptions whenever possible.
- Automatic database backup shall be performed periodically to prevent data loss.

2.5.3 Security

The HeatWatch AI system shall protect user information and system resources against unauthorized access and malicious activities.

Security requirements include:

- User passwords shall be securely hashed before storage.
- All communication between client and server shall use HTTPS encryption.
- API keys shall never be exposed to end users.
- Only authenticated users shall access protected system resources.
- Administrative functions shall be accessible only to authorized administrators.

- All user activities shall be recorded in secure audit logs.
- User input shall be validated to prevent SQL Injection, Cross-Site Scripting (XSS), and other common web vulnerabilities.
- Sensitive configuration files shall remain inaccessible to normal users.

2.5.4 Maintainability

The HeatWatch AI system shall be designed to simplify future enhancements and maintenance.

Maintainability requirements include:

- The software shall follow a modular architecture.
- Weather data collection, AI prediction, dashboard visualization, and report generation shall be implemented as independent modules.
- Source code shall follow consistent coding standards and naming conventions.
- All important modules shall include sufficient documentation.
- Configuration settings shall be stored separately from application logic.
- New machine learning models shall be integrated without requiring major architectural changes.
- The system shall support future addition of new weather APIs and data sources.

2.5.5 Portability

The software shall be capable of operating on multiple software platforms with minimal modifications.

Portability requirements include:

- The application shall support Windows and Linux operating systems.
- The web application shall function correctly on modern browsers including Google Chrome, Mozilla Firefox, Microsoft Edge, and Brave.
- The backend shall be developed using Python to ensure platform independence.
- Database migration between supported database systems shall require minimal changes.
- The application shall support deployment on both local servers and cloud platforms.

2.5.6 Performance

The HeatWatch AI system shall satisfy the following performance requirements.

Static Requirements

- The database shall support storage of historical weather records for multiple years.
- The application shall support at least 500 registered users.
- The system shall maintain weather and sentiment history for multiple geographical locations.
- Storage shall support future expansion without significant performance degradation.

Dynamic Requirements

- Dashboard loading time shall not exceed 5 seconds under normal conditions.
- Heatwave prediction shall be generated within 10 seconds after receiving new input data.
- API responses shall be processed in real time whenever possible.
- The system shall support multiple simultaneous users without noticeable reduction in performance.
- Historical reports shall be generated within 15 seconds for normal datasets.
- Alert notifications shall be generated immediately after a high-risk heatwave prediction.

2.6 Database Requirements

The HeatWatch AI system requires a relational database for storing user information, weather records, AI prediction results, alerts, reports, and historical datasets.

2.6.1 Database Entities

The major entities of the database are:

- Users
- Weather Data
- Social Media Posts

- Sentiment Analysis Results
- Heatwave Predictions
- Alerts
- Historical Reports
- System Logs

2.6.2 Entity Relationships

The relationships between the entities are summarized below.

- One User can generate multiple Reports.
- One Weather Record can have multiple Prediction Results.
- One Prediction can generate multiple Alerts.
- Multiple Social Media Posts contribute to one Sentiment Analysis Result.
- Sentiment Analysis Results and Weather Records together produce Heatwave Predictions.
- Each Prediction is associated with one geographical location and timestamp.

2.6.3 Database Constraints

The database shall satisfy the following integrity constraints.

- Every registered user shall possess a unique User ID.
- Every email address shall be unique.
- Weather records shall contain valid timestamps.
- Prediction records shall always reference an existing weather record.
- Foreign key constraints shall maintain referential integrity.
- Primary keys shall uniquely identify every database record.
- Database transactions shall preserve ACID properties.
- Backup copies shall be generated periodically to ensure disaster recovery.

2.6.4 Database Access Requirements

The database management system shall provide:

- Secure authentication for database access.
- Fast retrieval of weather and prediction records.
- Indexed search for historical reports.
- Concurrent access by multiple authenticated users.
- Backup and recovery mechanisms.
- Data encryption for sensitive user information.

2.7 Summary

The Software Requirement Specification defines both the functional and non-functional requirements of the proposed HeatWatch AI system. These requirements establish a clear foundation for system design, implementation, testing, and future maintenance. The specified requirements ensure that the software remains reliable, secure, scalable, maintainable, and capable of providing accurate heatwave monitoring, prediction, and early warning services through the integration of meteorological data and Artificial Intelligence.

Chapter 3

Software Project Management Plan (SPMP)

3.1 Introduction

3.1.1 Project Overview

The software being developed is HeatWatch AI, an AI-Based Climate Intelligence System for Heatwave Monitoring, Prediction, and Early Warning. The primary purpose of this software is to integrate meteorological data with Artificial Intelligence-based social media sentiment analysis to provide timely heatwave alerts. The project is executed for disaster management officers, government authorities, environmental researchers, and system administrators to support timely decision-making.

3.1.2 Project Deliverables

The system will generate the following project deliverables during its life span:

- **Source Code:** Python backend code, Flask/Django framework files, and Machine Learning models utilizing Scikit-learn and TensorFlow.
- **Databases:** MySQL database schema and records containing weather data, user data, and sentiment results.
- **Documents:** Software Requirement Specification (SRS) and Historical PDF/CSV Reports.
- **Executable Code:** Fully functional web-based platform accessible via modern browsers.

3.2 Project Organization

3.2.1 Software Process Model

The project follows the Agile Scrum software development model.

Agile is the most suitable process model because AI applications continuously evolve with new datasets, changing user requirements and model improvements. The project requires frequent testing of machine learning models, integration of new APIs and regular feedback from users. Scrum allows development in short iterations, making it easy to improve prediction accuracy, dashboard features and alert mechanisms while maintaining software quality.

3.2.2 Roles and Responsibilities

Role	Responsibilities
Project Manager	Planning, scheduling, resource allocation and project monitoring
Business Analyst	Requirement gathering and stakeholder communication
UI/UX Designer	Designing wireframes, dashboard and user interface
Backend Developer	API integration, database management and business logic
AI/ML Engineer	Sentiment analysis model and heatwave prediction model
Frontend Developer	Dashboard implementation and user interaction
Database Administrator	Database design, optimization and security
Tester (QA Engineer)	Functional testing, bug reporting and performance testing

Table 3.1: Project Roles and Responsibilities

3.2.3 Tools and Techniques

- **Programming Languages:** Python 3.x for the backend to ensure platform independence.
- **Frameworks:** Flask or Django for web application services.
- **Databases:** MySQL for storing user information, weather data, and alerts.

- **APIs:** OpenWeatherMap (or similar) for meteorological data, and Social Media APIs for retrieving public posts.
- **Machine Learning:** Scikit-learn and TensorFlow for sentiment classification and prediction.
- **Visualizations:** Matplotlib or Plotly for generating interactive charts and graphs on the dashboard.

3.3 Project Management Plan

3.3.1 Tasks

SPMP-T0001: Requirement Analysis

Description: Gather project requirements, identify stakeholders, define project objectives, and analyse functional and non-functional requirements.

Deliverables and Milestones: Requirement Specification Document. Milestone: Requirements Finalized (21 July 2026).

Resources Needed: Project team, stakeholder discussions, documentation tools.

Dependencies and Constraints: Initial project activity; depends on stakeholder availability.

Risks and Contingencies: Requirement changes may occur; conduct regular review meetings.

SPMP-T0002: Software Requirement Specification (SRS)

Description: Prepare the detailed Software Requirement Specification (SRS) document describing all system functionalities, interfaces, constraints and requirements.

Deliverables and Milestones: Completed Software Requirement Specification (SRS). Milestone: SRS Completed (07 August 2026).

Resources Needed: Documentation software, requirement analysis results.

Dependencies and Constraints: Depends on completion of Requirement Analysis.

Risks and Contingencies: Incomplete requirements may require revisions before approval.

SPMP-T0003: Software Project Management Plan (SPMP)

Description: Prepare the Software Project Management Plan (SPMP) covering project scope, schedule, resources, milestones and risk management.

Deliverables and Milestones: Completed SPMP Document. Milestone: SPMP Completed (07 August 2026).

Resources Needed: Project planning tools and documentation software.

Dependencies and Constraints: Depends on finalized project requirements.

Risks and Contingencies: Changes in project scope may require updates to the management plan.

SPMP-T0004: Database Design

Description: Design the relational database schema for storing weather information, user data, alerts and sentiment analysis results.

Deliverables and Milestones: Database schema and Entity Relationship Diagram (ERD).

Resources Needed: MySQL and database modelling tools.

Dependencies and Constraints: Depends on completion of SRS.

Risks and Contingencies: Future system enhancements may require database restructuring.

SPMP-T0005: UI/UX Design

Description: Design user-friendly interfaces, dashboard layouts and application workflows.

Deliverables and Milestones: Wireframes, UI mock-ups and interface prototype.

Resources Needed: Figma or similar UI design software.

Dependencies and Constraints: Depends on finalized requirements.

Risks and Contingencies: User feedback may require interface modifications.

SPMP-T0006: Weather API Integration

Description: Integrate weather APIs to retrieve real-time meteorological information required for prediction.

Deliverables and Milestones: Functional Weather API Module. Milestone: APIs Integrated (25 August 2026).

Resources Needed: OpenWeatherMap API, Python environment and internet connectivity.

Dependencies and Constraints: Requires valid API keys and stable internet access.

Risks and Contingencies: API downtime may require cached historical weather data.

SPMP-T0007: Social Media Data Collection Module

Description: Develop modules to collect publicly available social media posts related to heatwave events.

Deliverables and Milestones: Social Media Data Collection Module. Milestone: APIs

Integrated (25 August 2026).

Resources Needed: Social Media APIs, Python libraries and internet access.

Dependencies and Constraints: Subject to API rate limits and privacy regulations.

Risks and Contingencies: Alternative data sources will be used if API services become unavailable.

SPMP-T0008: Machine Learning Model Development

Description: Develop and train machine learning models for predicting heatwave severity using historical weather and social media datasets.

Deliverables and Milestones: Heatwave Prediction Model. Milestone: AI Modules Completed (15 September 2026).

Resources Needed: Scikit-learn, TensorFlow and training datasets.

Dependencies and Constraints: Depends on successful completion of data collection.

Risks and Contingencies: Poor prediction accuracy may require additional feature engineering and model retraining.

SPMP-T0009: AI Sentiment Analysis Module

Description: Develop Natural Language Processing (NLP) models to analyse public sentiment regarding heatwave conditions.

Deliverables and Milestones: Sentiment Analysis Module. Milestone: AI Modules Completed (15 September 2026).

Resources Needed: NLP libraries and labelled datasets.

Dependencies and Constraints: Depends on collected social media data.

Risks and Contingencies: Noisy textual data may reduce model accuracy and require preprocessing.

SPMP-T0010: Authentication & User Management

Description: Develop secure user registration, login, authentication and user profile management features.

Deliverables and Milestones: Authentication and User Management Module.

Resources Needed: Flask/Django authentication framework and MySQL database.

Dependencies and Constraints: Depends on database implementation.

Risks and Contingencies: Implement password hashing and secure authentication mechanisms.

SPMP-T0011: Dashboard Development

Description: Develop the interactive dashboard for displaying weather information, AI predictions, alerts and visual analytics.

Deliverables and Milestones: Interactive Dashboard. Milestone: Dashboard Completed (29 September 2026).

Resources Needed: HTML, CSS, JavaScript, Flask/Django and visualization libraries.

Dependencies and Constraints: Depends on backend APIs and AI modules.

Risks and Contingencies: Performance optimization may be required for large datasets.

SPMP-T0012: Alert Generation Module

Description: Develop an automated alert generation system for predicted heatwave events.

Deliverables and Milestones: Alert Generation Module.

Resources Needed: Backend services and notification APIs.

Dependencies and Constraints: Depends on AI prediction results.

Risks and Contingencies: False alerts will be minimized through threshold optimization.

SPMP-T0013: Report Generation Module

Description: Develop modules to generate downloadable PDF and CSV reports containing weather trends and prediction analysis.

Deliverables and Milestones: Report Generation Module.

Resources Needed: PDF libraries and reporting framework.

Dependencies and Constraints: Depends on dashboard and database modules.

Risks and Contingencies: Validate report accuracy before report generation.

SPMP-T0014: System Integration

Description: Integrate frontend, backend, database, APIs and AI modules into a unified software system.

Deliverables and Milestones: Fully Integrated System. Milestone: System Integration Completed (27 October 2026).

Resources Needed: Development environment and version control system.

Dependencies and Constraints: Depends on successful completion of all development modules.

Risks and Contingencies: Integration conflicts will be resolved through debugging and testing.

SPMP-T0015: Unit Testing

Description: Perform unit testing to verify the functionality of individual software modules.

Deliverables and Milestones: Unit Testing Report.

Resources Needed: Software testing tools and test cases.

Dependencies and Constraints: Depends on completion of individual modules.

Risks and Contingencies: Defects will be corrected before integration testing.

SPMP-T0016: Integration & System Testing

Description: Perform integration testing and complete end-to-end system testing to validate the complete application.

Deliverables and Milestones: System Testing Report. Milestone: Testing Completed (17 November 2026).

Resources Needed: Testing tools, datasets and QA environment.

Dependencies and Constraints: Depends on successful system integration.

Risks and Contingencies: Compatibility issues will be resolved before deployment.

SPMP-T0017: Bug Fixing & Performance Optimization

Description: Resolve identified software defects and optimize system performance for reliability and efficiency.

Deliverables and Milestones: Optimized Stable System.

Resources Needed: Debugging and performance profiling tools.

Dependencies and Constraints: Depends on testing reports.

Risks and Contingencies: Regression testing will be performed after every major fix.

SPMP-T0018: Documentation & Final Report

Description: Prepare the complete technical documentation, user manual and final project report.

Deliverables and Milestones: Final Project Documentation. Milestone: Documentation Completed (28 November 2026).

Resources Needed: Documentation software and project records.

Dependencies and Constraints: Depends on completion of development and testing.

Risks and Contingencies: Documents will undergo multiple reviews before final submission.

SPMP-T0019: Final Review, Deployment & Presentation

Description: Conduct the final project review, deploy the completed system and deliver the final presentation.

Deliverables and Milestones: Final Deployment and Project Presentation. Milestone: Final Submission & Presentation (01 December 2026).

Resources Needed: Deployment server, presentation slides and project documentation.

Dependencies and Constraints: Depends on successful testing and documentation completion.

Risks and Contingencies: Maintain deployment backup and presentation contingency plans in case of unexpected issues.

3.3.2 Assignments

- **Vedang Kulkarni (16014224053):** Assigned to Requirement Analysis, Software Requirement Specification (SRS), Software Project Management Plan (SPMP), Database Design, Weather API Integration, Authentication & User Management, System Integration, Integration & System Testing, Bug Fixing & Performance Optimization, Documentation & Final Report, and Final Review, Deployment & Presentation.
- **Rajat Kumar (16014224054):** Assigned to UI/UX Design, Social Media Data Collection Module, Machine Learning Model Development, AI Sentiment Analysis Module, Dashboard Development, Alert Generation Module, Report Generation Module, Unit Testing, and support during System Integration and Final Deployment.

3.3.3 Timetable

The project timetable showing the estimated start and completion dates for each task is best illustrated using a Gantt chart diagram. This timetable is updated throughout the project life span to reflect changes in the execution time for tasks and completion dates.

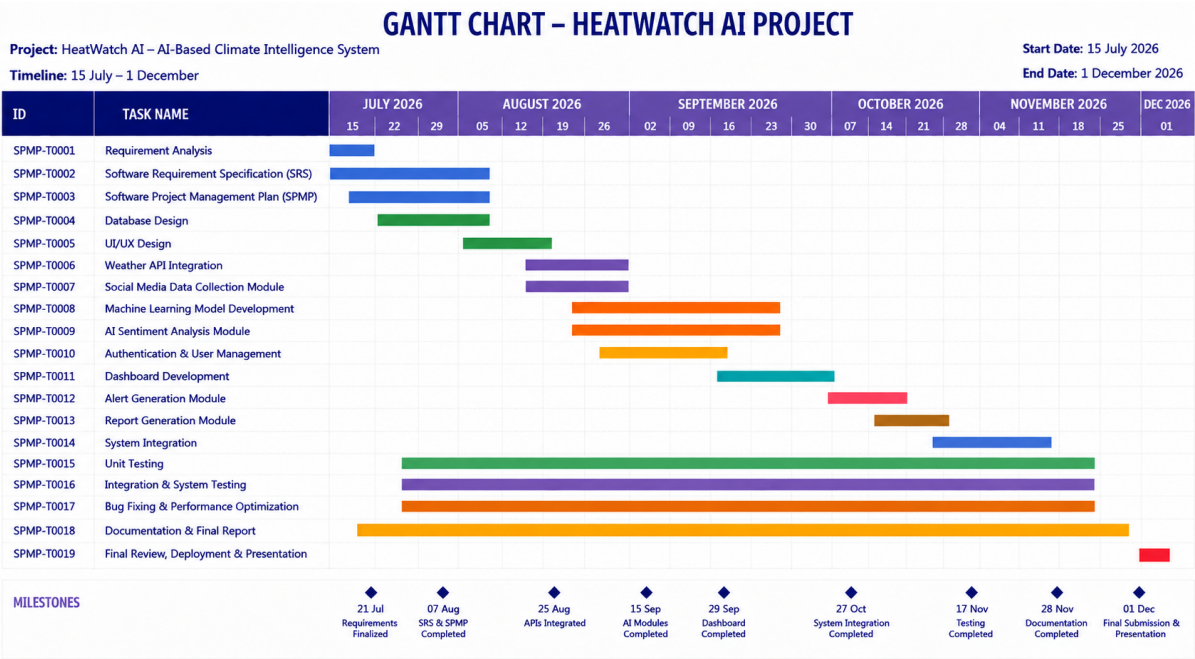


Figure 3.1: Gantt Chart for HeatWatch AI Project

Chapter 4

Behavioural UML Diagrams

Behavioural UML diagrams represent the dynamic behaviour of the HeatWatch AI system. They describe user interactions, system workflows, communication between components, and changes in system states during execution. The following diagrams model the major behavioural aspects of the proposed system.

4.1 Use Case Diagram

The Use Case Diagram represents the interaction between different actors and the major functionalities provided by the HeatWatch AI system.

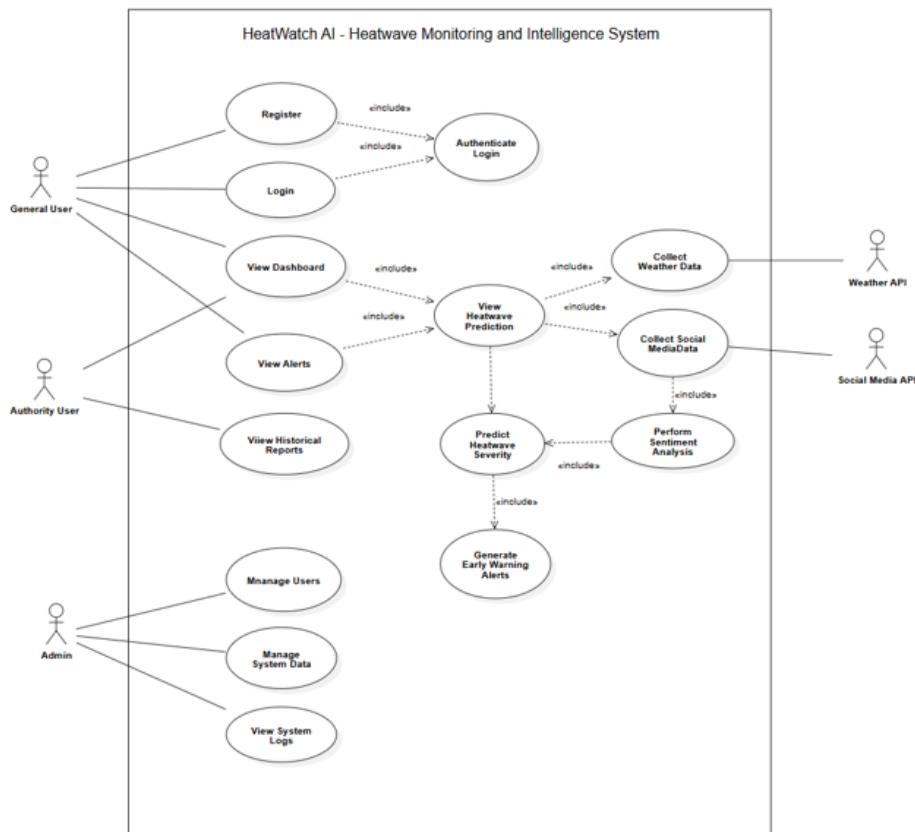


Figure 4.1: Use Case Diagram for HeatWatch AI

4.2 Activity Diagrams

Activity diagrams represent the workflow of important processes performed by the Heat-Watch AI system, including actions, decisions and transitions between activities.

4.2.1 User Registration and Login Activity

This activity diagram represents the workflow for user registration, authentication, validation of credentials and successful redirection to the system dashboard.

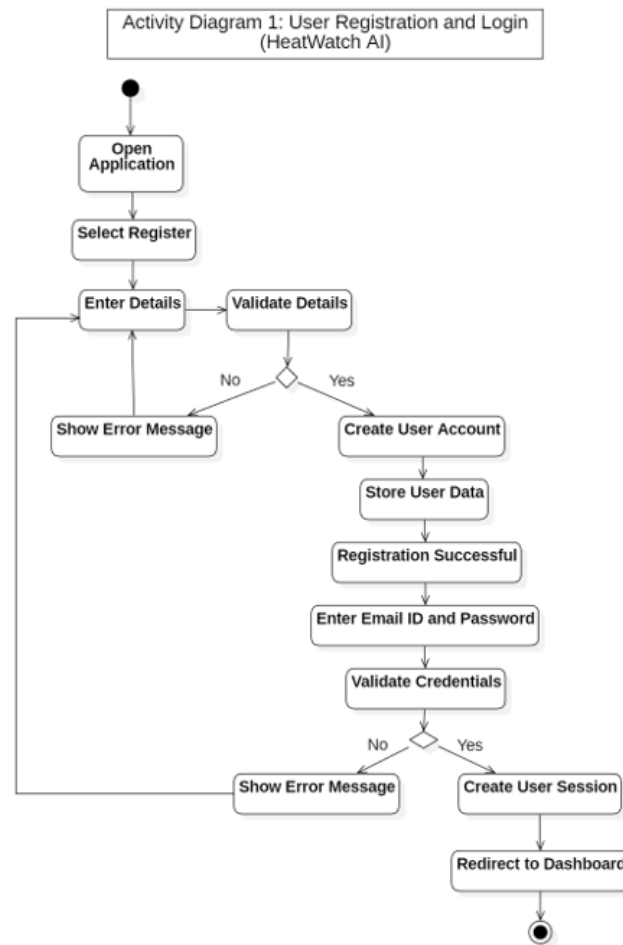


Figure 4.2: Activity Diagram for User Registration and Login

4.2.2 Heatwave Monitoring and Alert Generation Activity

This activity diagram represents the workflow for weather and social-media data collection, preprocessing, sentiment analysis, heatwave severity prediction and generation of early warning alerts.

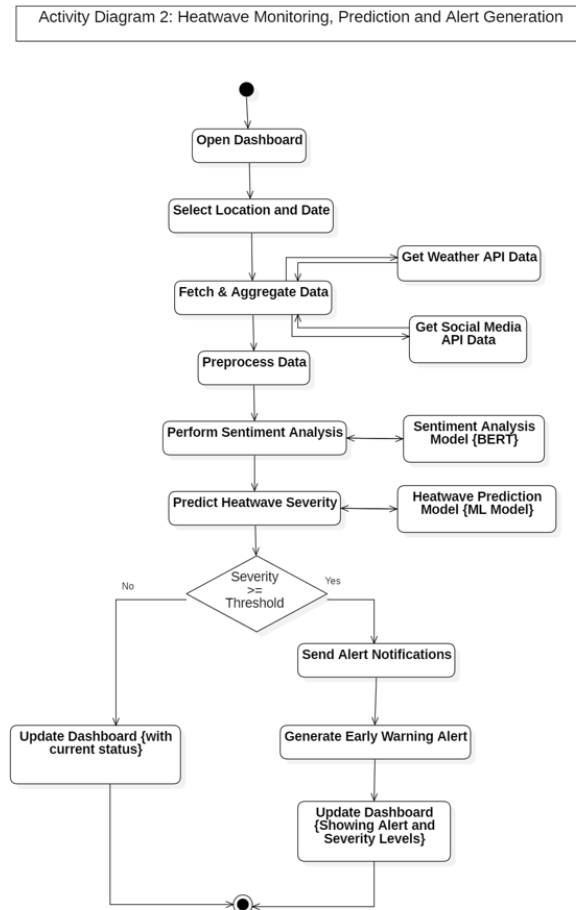


Figure 4.3: Activity Diagram for Heatwave Monitoring, Prediction and Alert Generation

4.3 Sequence Diagram

The Sequence Diagram represents the time-ordered interaction between the authorised user, web dashboard, AI engine and external APIs during heatwave monitoring, prediction and alert generation.

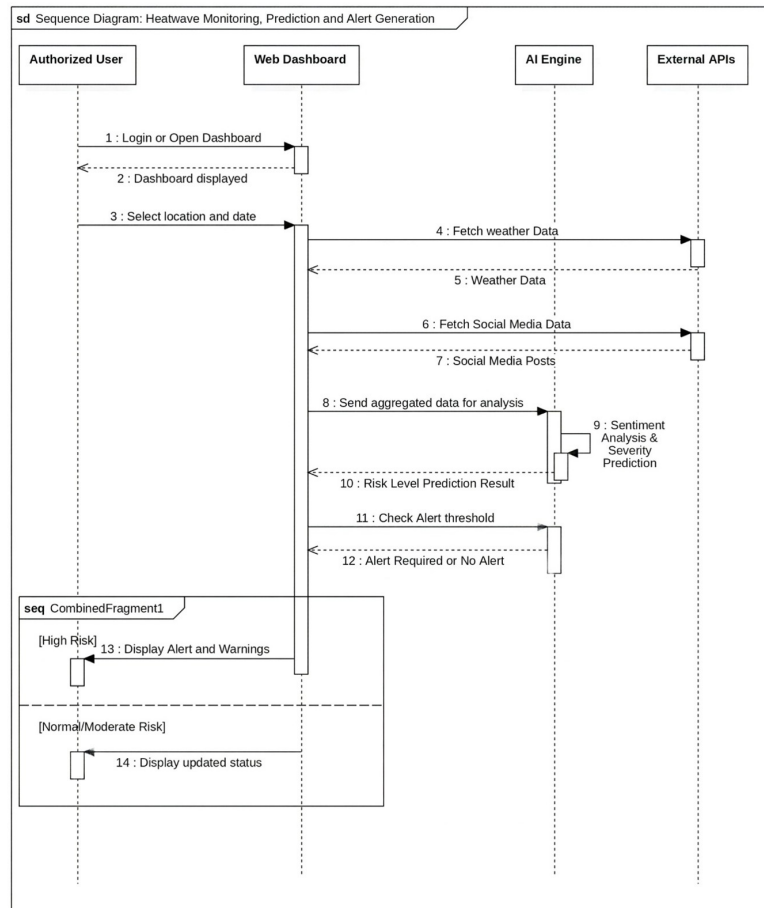


Figure 4.4: Sequence Diagram for HeatWatch AI

4.4 State Machine Diagram

The State Machine Diagram represents the major states of the HeatWatch AI system and the events responsible for transitions between these states during system operation.

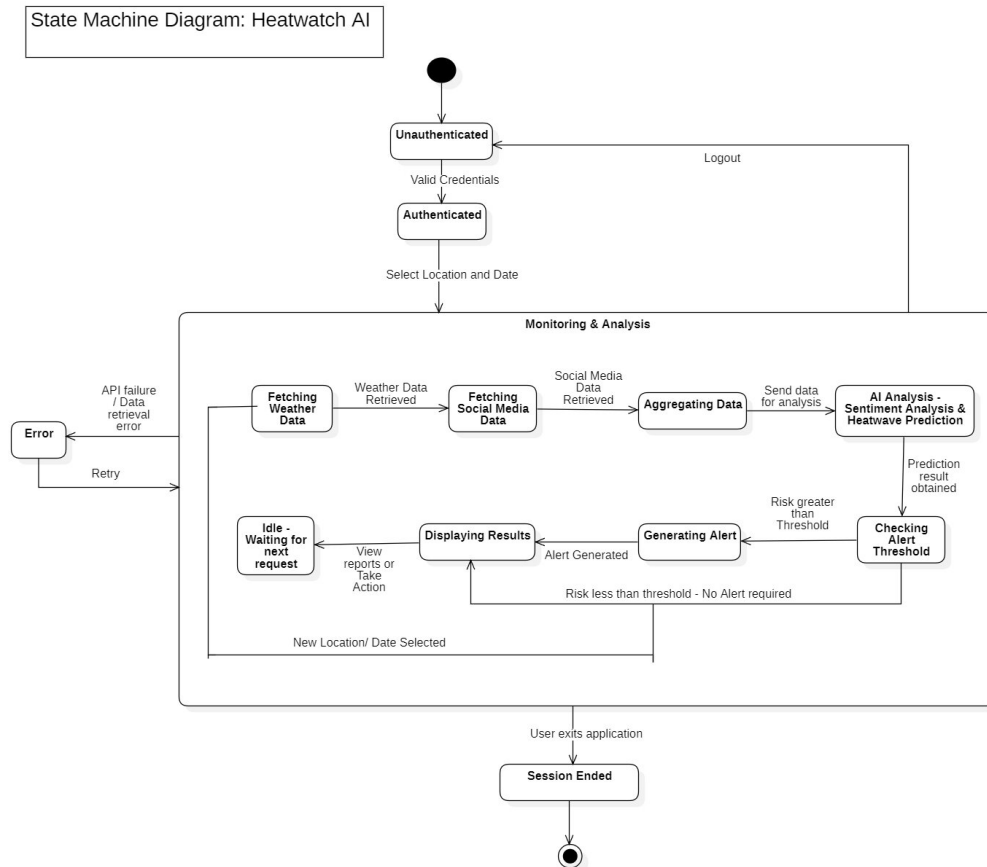


Figure 4.5: State Machine Diagram for HeatWatch AI

4.5 Collaboration Diagram

The Collaboration Diagram, also known as a Communication Diagram, represents the communication between the major objects and components of the HeatWatch AI system through numbered message exchanges.

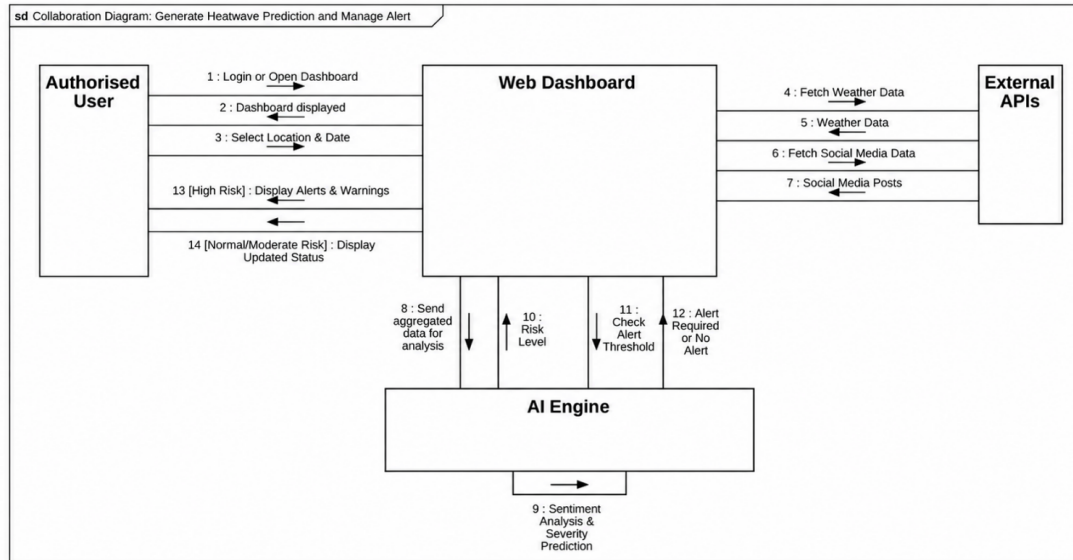


Figure 4.6: Collaboration Diagram for HeatWatch AI

4.6 Summary

The behavioural UML diagrams describe the dynamic operation of the HeatWatch AI system. The Use Case Diagram represents user interactions, the Activity Diagrams describe major system workflows, the Sequence Diagram shows time-ordered communication, the State Machine Diagram represents system state transitions, and the Collaboration Diagram describes message-based interaction among system components. Together, these diagrams provide a clear behavioural model for the design and implementation of HeatWatch AI.